



Snowflake Real-Time Query Alerts

Architecture · long-running, queued and lock-blocked queries · E-mail, Microsoft Teams, Slack, PagerDuty

v2.0 · OBSERVABILITY.QUERY_ALERTS · October 2026

What the Admin team is alerted on



15 min

LONG_RUNNING

Query executing 15 min or more. Time spent queued or waiting on a lock is excluded.

Status RUNNING



5 min

QUEUED

Query waiting for warehouse capacity or a warehouse resume for 5 min or more.

Status QUEUED, RESUMING_WAREHOUSE



5 min

BLOCKED

Query waiting on a table lock for 5 min or more. Names the locked table and the blocking query.

Status BLOCKED

about 1 min

detection delay

4

notification channels

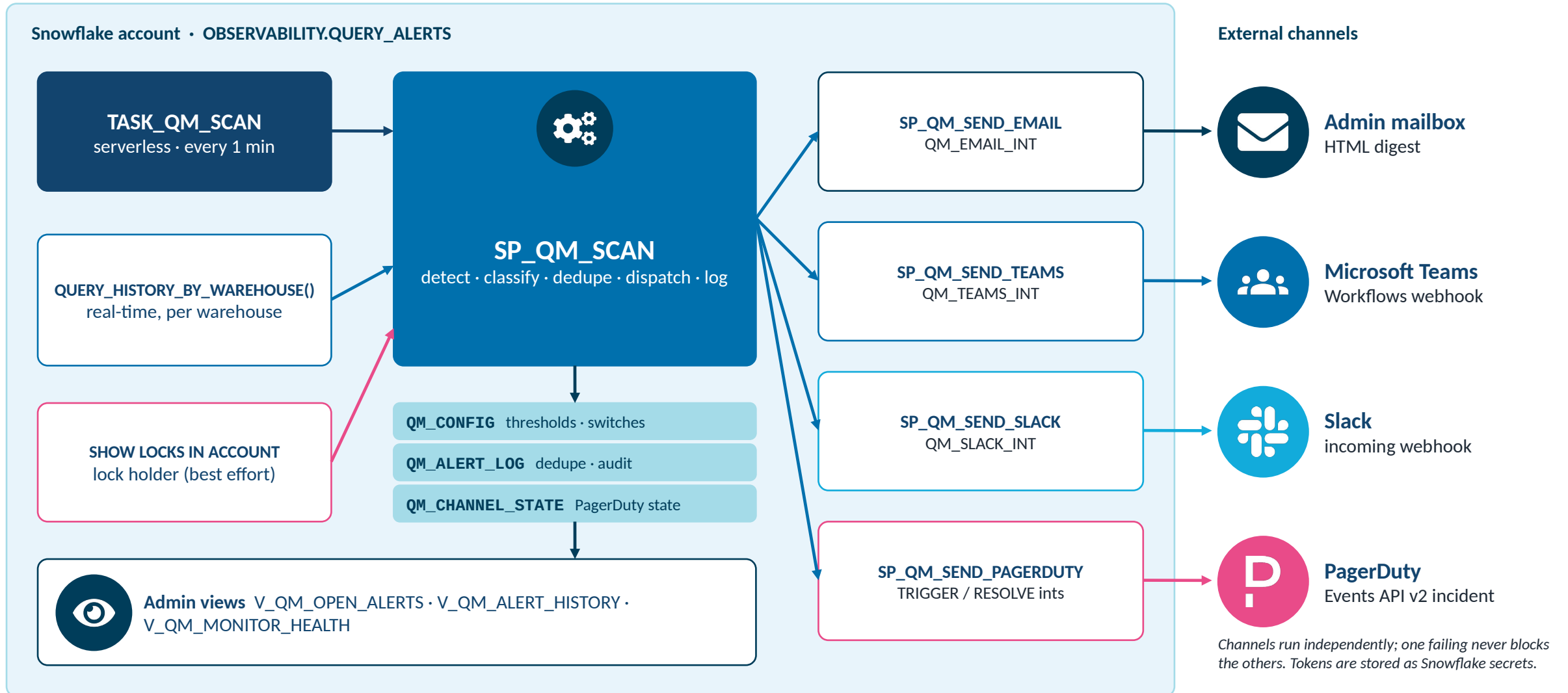
0

external servers

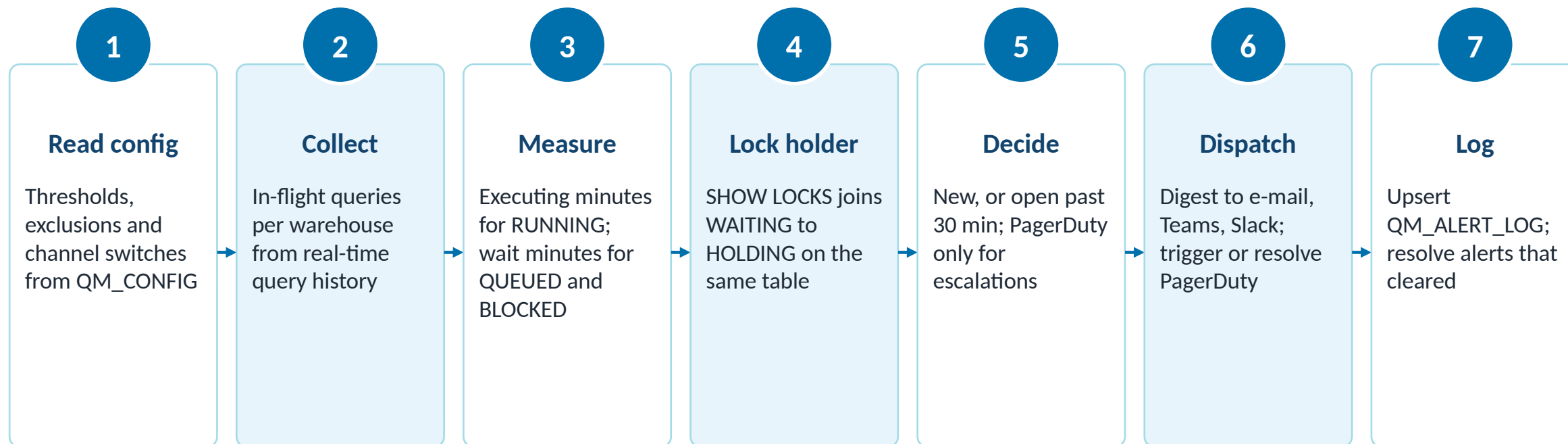
30 min

re-notify window

Solution architecture



Processing flow, every minute



Executing minutes elapsed since start minus queued, provisioning, repair and lock time. **Re-notify** the same query and alert type is repeated at most every 30 min while it stays open. **Isolation** each channel call has its own error handler.

Notification channels



E-mail

PURPOSE

Admin list, audit trail

FORMAT

HTML table with action SQL

VOLUME

1 digest per run, max 25 rows

SETUP

Verified Snowflake user e-mails



Microsoft Teams

PURPOSE

Admin channel awareness

FORMAT

Adaptive Card list

VOLUME

Same digest as e-mail

SETUP

Workflows webhook URL, id as secret



Slack

PURPOSE

Admin channel, if Slack is used

FORMAT

mrkdwn bullet list

VOLUME

Same digest as e-mail

SETUP

Incoming webhook, path as secret



PagerDuty

PURPOSE

On-call escalation only

FORMAT

Events API v2, severity error

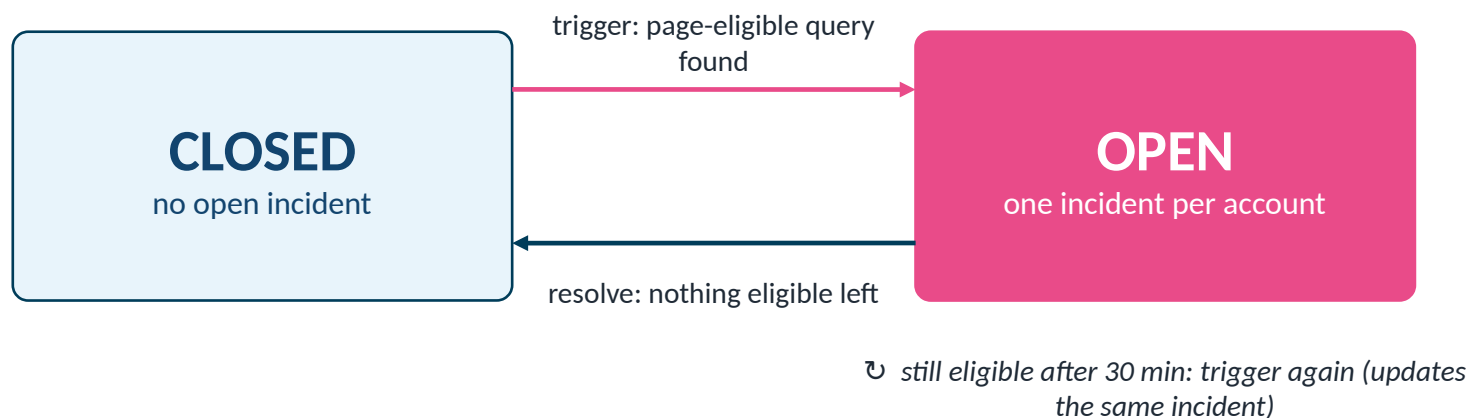
VOLUME

1 incident per account, auto-resolve

SETUP

Events v2 routing key as secret

PagerDuty: escalation and auto-resolve



Two integrations, one secret. Snowflake webhook templates are static, so QM_PAGERDUTY_TRIGGER_INT and QM_PAGERDUTY_RESOLVE_INT share the routing key and a fixed dedup_key. Incident state is tracked in QM_CHANNEL_STATE. Give each Snowflake account its own dedup_key.

Paging policy

TYPES

LONG_RUNNING, BLOCKED

ESCALATION

query at 30 min or more

RE-TRIGGER

every 30 min while eligible

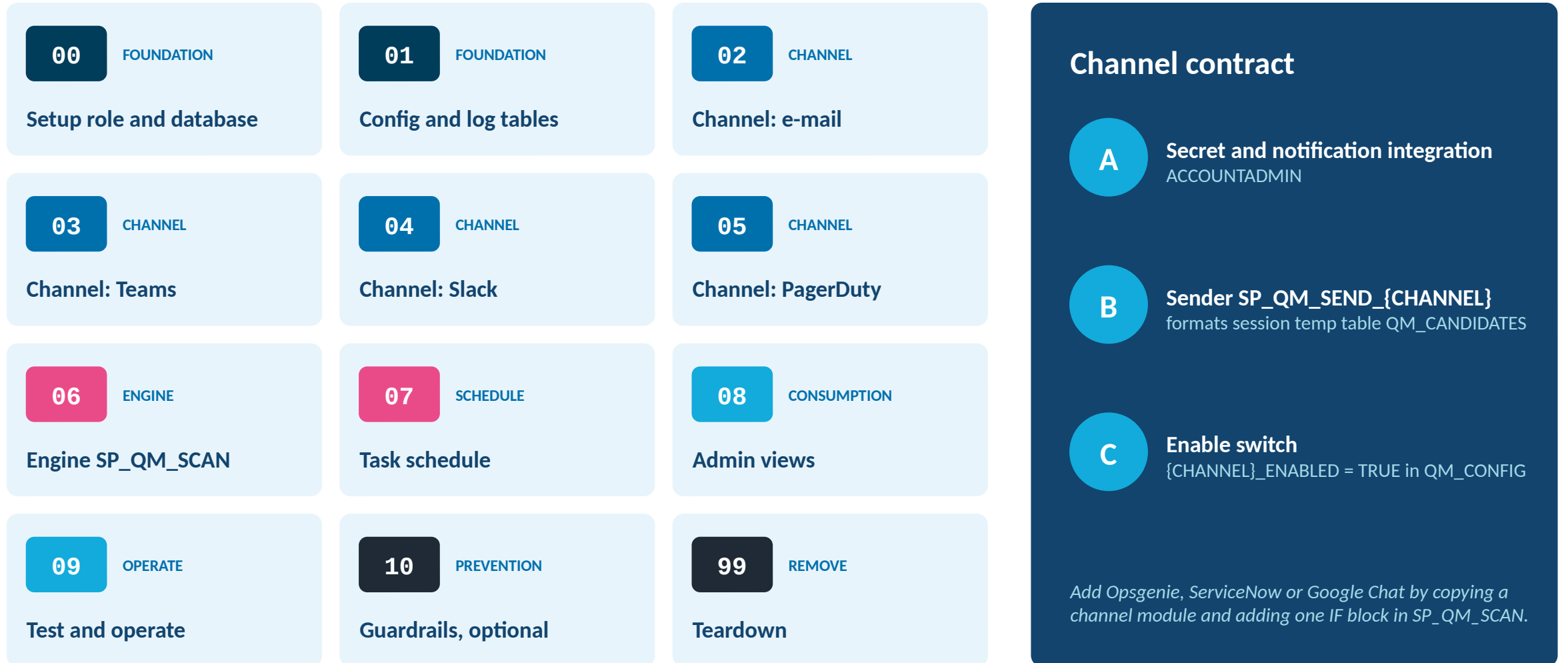
SEVERITY

error

AUTO-RESOLVE

on, configurable

Modular code: one script per concern



Security and privileges

Need	Grant to OPS_QUERY_MONITOR	Note
See all users' queries in real time	MONITOR on every warehouse	Re-run the grant block when a warehouse is created
Run the serverless task	EXECUTE TASK, EXECUTE MANAGED TASK	No warehouse to manage
Send notifications	USAGE on integrations and secrets	Tokens never appear in code or logs
Identify the lock holder	SHOW LOCKS IN ACCOUNT needs ACCOUNTADMIN	Without it, blocked queries still alert, no holder id
Setup	ACCOUNTADMIN, once	Scripts 00 and 02 to 05, section A



Privacy by default

- Query text excluded from messages (INCLUDE_QUERY_TEXT = FALSE)
- Webhook tokens stored as Snowflake secrets
- E-mail only to verified Snowflake users
- No IMPORTED PRIVILEGES on the SNOWFLAKE database

Design decisions and limitations

Topic	Decision	Why
Real-time source	INFORMATION_SCHEMA query history	ACCOUNT_USAGE.QUERY_HISTORY lags up to about 45 min
Polling	Serverless task every minute	No native event for a query passing N minutes
Task, not alert object	Loop over warehouses, multi-channel dispatch	An alert condition must be a single query
Per-warehouse scan	Up to 10,000 recent queries each	A busy warehouse cannot hide another's query
Alert fatigue	Digest, 30 min re-notify, exclusions	PagerDuty only for escalations
Cost	One XSMALL serverless run per minute	Typically seconds of compute; 2-minute schedule halves it
Known limit	Warehouse-less queries not covered	Metadata-only statements rarely run long

Deploy and operate

1

Edit placeholders

Recipients, Teams URL, Slack path, PagerDuty key

2

Deploy in order

deploy/deploy_all.sql with SnowSQL, or Snowflake CLI

3

Test channels

CALL SP_QM_TEST_CHANNELS()

4

Test end to end

SYSTEM\$WAIT and a two-session lock

5

Start schedule

Resume TASK_QM_SCAN



Day-2 operations

```
SELECT * FROM V_QM_OPEN_ALERTS;  
SELECT * FROM V_QM_MONITOR_HEALTH;  
SELECT SYSTEM$CANCEL_QUERY('{query_id}');  
SELECT SYSTEM$ABORT_TRANSACTION({txn_id});  
UPDATE QM_CONFIG SET PARAM_VALUE = '20'  
    WHERE PARAM_NAME = 'LONG_RUNNING_MIN';
```

Runbook: re-run the MONITOR grant for new warehouses. The task auto-suspends after 10 consecutive failures; review monitor health weekly.